

In this exercise, you'll add user registration and authentication to your chatbot application. Currently, anyone can access your chatbot without restrictions. For enterprise-grade applications, proper user management is essential for security, personalization, and compliance.

What You'll Learn

By the end of this exercise, you will have:

- Implemented user registration and login functionality using Amazon Cognito
- Protected your chatbot API with authentication requirements
- Created a seamless user experience with login/logout flows
- Learned to integrate AWS Cognito with your existing application
- Enhanced security by requiring authentication before chatbot access

Step 1: Implement User Authentication System

Instruct Kiro to add comprehensive user authentication to your chatbot application using Amazon Cognito.

Implement user registration and authentication for the chatbot.

- Use Cognito for user management.
- Only require an email address and a password in the registration step.
- Users must login before they can chat with the chatbot.

Use "Project=kiro-workshop" tag for all new resources created.

Enterprise Security

User authentication is a critical component of enterprise applications. This exercise demonstrates how to quickly implement production-ready authentication using AWS managed services.

Expected Authentication Components

Kiro should implement:

Frontend Changes

- Login/Register forms with email and password fields
- User session management and token storage
- Protected routes that require authentication
- Logout functionality with session cleanup
- User-friendly error handling for auth failures
- Responsive design that works on all devices

Step 2: Test the Authentication Flow

Once Kiro implements the authentication system, thoroughly test the complete user experience by registering a new user, logging in, and verifying that only authenticated users can access the chatbot.

Authentication Testing Checklist

Verify these authentication scenarios work properly:

- ☐ **User Registration:** New users can create accounts with email/password
- ☐ **Email Verification:** Users receive and can confirm verification emails
- ☐ **User Login:** Registered users can log in successfully
- ☐ **Chatbot Access:** Authenticated users can use the chatbot normally
- ☐ **Access Protection:** Unauthenticated users cannot access chatbot features
- ☐ **Session Management:** User sessions persist across browser refreshes
- ☐ **Logout:** Users can log out and session is properly cleared
- ☐ **Error Handling:** Clear error messages for invalid credentials

Step 3: Validate Security and User Experience

Ensure the authentication implementation meets security best practices and provides a smooth user experience.

Please verify that the authentication is working properly and that unauthenticated users cannot access the chatbot API.

Security Validation

Authentication systems require thorough testing to ensure security. Pay special attention to token validation, session management, and proper error handling.

Expected Results

After completing this exercise, you should have:

- ☒ **User Registration:** New users can create accounts with email verification
- ☒ **Secure Login:** Authenticated access using JWT tokens
- ☒ **Protected Chatbot:** Only authenticated users can access chat functionality
- ☒ **Session Management:** Proper token storage and session handling
- ☒ **User Experience:** Intuitive login/register interface
- ☒ **Security:** Backend API properly validates authentication tokens
- ☒ **Error Handling:** Clear feedback for authentication failures

Troubleshooting Authentication Issues

► Cognito Configuration Errors

► JWT Token Validation Fails

► CORS Issues with Authentication

► Email Verification Not Working

► Frontend Authentication State Issues

Architecture Overview

► View Authenticated Application Architecture

Key Authentication Features

Your enhanced application now includes:

User Management

- **Amazon Cognito Integration:** Managed user authentication service
- **Email Verification:** Secure account activation process
- **Password Security:** AWS-managed password policies and encryption
- **JWT Tokens:** Industry-standard authentication tokens
- **Session Management:** Secure token storage and refresh handling

Security Enhancements

- **API Protection:** Backend validates all requests require authentication
- **Token Validation:** JWT signatures verified against Cognito
- **Secure Storage:** Authentication tokens stored securely in browser
- **HTTPS Only:** All authentication traffic encrypted in transit
- **Error Handling:** Secure error messages that don't leak information

User Experience

- **Seamless Integration:** Authentication flows integrated into existing UI
- **Responsive Design:** Login/register forms work on all devices
- **Clear Feedback:** User-friendly messages for all authentication states
- **Persistent Sessions:** Users stay logged in across browser sessions
- **Easy Logout:** Simple logout process that clears all session data

Congratulations! 🎉

Outstanding achievement! You've successfully built a complete, enterprise-grade AI chatbot application with user authentication. Your chatbot now has secure user registration and login functionality, ensuring only authenticated users can access the AI-powered chat features.

Your Incredible Journey

Think about how far you've come since Exercise 1:

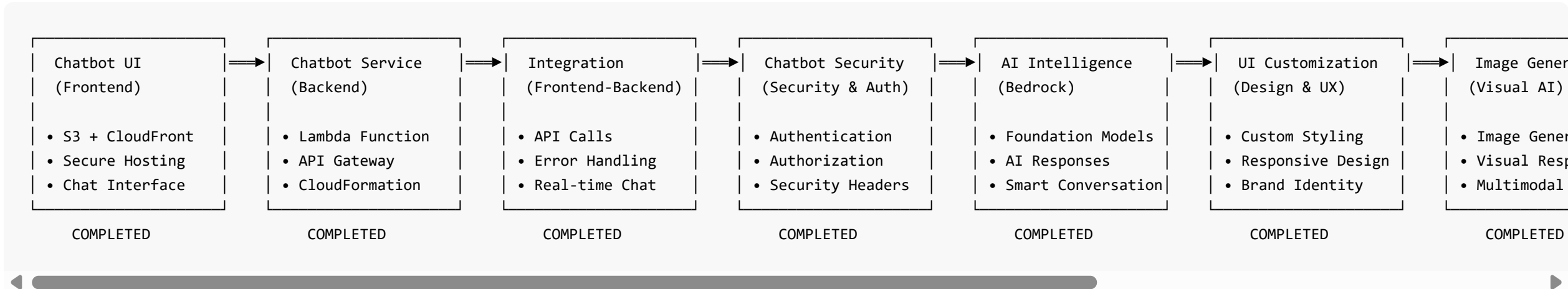
- **Started with:** A simple idea to build a chatbot
- **Built:** A secure, scalable, AI-powered application with enterprise authentication
- **Learned:** Modern serverless architecture, AI integration, and security best practices
- **Achieved:** A production-ready chatbot that rivals commercial solutions

From a basic HTML page to a sophisticated application with:

- ☒ Secure hosting on AWS with global CDN
- ☒ Serverless backend with auto-scaling
- ☒ AI intelligence powered by Amazon Bedrock
- ☒ Custom UI with professional design
- ☒ Multimodal capabilities (text, images, voice)
- ☒ Enterprise-grade user authentication
- ☒ Security hardening and best practices

You've transformed from having just an idea into building a complete, enterprise-ready AI application using cutting-edge AWS services. This is exactly the kind of innovation that drives modern software development!

Your Complete Application Architecture



You've built something truly remarkable! Your application demonstrates mastery of modern cloud architecture, AI integration, and enterprise security practices. The next phase will focus on packaging your entire solution into a reusable CDK template and performing a clean deployment process.